

Hilbert-Generated Spaces via Invariant ℓ_2 Generators

Run `fa_banach_001`, target arXiv:2110.02320

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Source Problem

The source paper is C. Correa, M. Cuth, and J. Somaglia, *Characterizations of weakly $\mathcal{K}$ -analytic and Vasak spaces using projectional skeletons and separable PRI*, arXiv:2110.02320. Problem 34, on page 28, asks:

To characterize the class of Hilbert generated spaces using projectional skeletons and SPRI.

of Plichko spaces that were considered in [14]. For instance, we suggest the following.

Problem 34. *To characterize the class of Hilbert generated spaces using projectional skeletons and SPRI.*

Definitions

Call a set $A \subset X$ an ℓ_2 -generator for a Banach space X if $\overline{\text{span}}(A) = X$ and there is a constant $C < \infty$ such that, for every $x^* \in X^*$,

$$\sup \left\{ \sum_{a \in F} |x^*(a)|^2 : F \subset A \text{ finite} \right\} \leq C^2 \|x^*\|^2. \quad (1)$$

This finite-subset formulation avoids any ambiguity when A is uncountable.

A Banach space is Hilbert generated if it is the range closure of a bounded linear operator from a Hilbert space.

Proof Intuition

The Hilbert-generated condition is exactly the assertion that the space has a dense family whose coordinates are uniformly square-summable against every functional. If $T : H \rightarrow X$ has dense range and (e_i) is an orthonormal basis of H , then $A = \{Te_i\}$ is dense and

$$\sum_i |x^*(Te_i)|^2 = \|T^*x^*\|_H^2 \leq \|T\|^2 \|x^*\|^2.$$

Thus A countably supports X^* . This is precisely the kind of set on which the source paper's skeleton-to-SPRI construction can be made coordinatewise: after passing to a suitable subskeleton, each projection either keeps or kills each $a \in A$. Conversely, if such an A is present, the map $e_a \mapsto a$ from $\ell_2(A)$ is bounded by Cauchy-Schwarz and has dense range.

Theorem 1. *Let X be a nonseparable Banach space and put $\kappa = \text{dens}(X)$. The following are equivalent.*

- (i) X is Hilbert generated.
- (ii) X admits a projectional skeleton $(P_s)_{s \in \Gamma}$ and an ℓ_2 -generator $A \subset X$ such that

$$P_s a \in \{0, a\} \quad (a \in A, s \in \Gamma).$$

- (iii) X admits a SPRI $(Q_\alpha)_{\alpha \leq \kappa}$ and an ℓ_2 -generator $A \subset X$ such that

$$Q_\alpha a \in \{0, a\} \quad (a \in A, \alpha \leq \kappa).$$

Proof. Assume first that X is Hilbert generated. Let $T : H \rightarrow X$ be a bounded operator from a Hilbert space with dense range. Choose an orthonormal basis $(e_i)_{i \in I}$ of H and put $A = \{Te_i : i \in I, Te_i \neq 0\}$. Then $\overline{\text{span}}(A) = X$. For every finite $F \subset A$ choose indices $i(a)$ with $a = Te_{i(a)}$. For $x^* \in X^*$,

$$\sum_{a \in F} |x^*(a)|^2 \leq \sum_{i \in I} |x^*(Te_i)|^2 = \|T^* x^*\|_H^2 \leq \|T\|^2 \|x^*\|^2.$$

Thus A is an ℓ_2 -generator.

The same estimate implies that A countably supports every $x^* \in X^*$: for fixed $m \geq 1$, the set $\{a \in A : |x^*(a)| \geq 1/m\}$ is finite, and the support is the union of these finite sets over m . Since X^* is norming, X is Plichko. By the standard equivalence quoted in the source paper, Plichko spaces admit commutative projectional skeletons. Applying Corollary 21 of the source paper to this skeleton and to the set A , we may pass to an isomorphic subskeleton satisfying $P_s a \in \{0, a\}$ for all $a \in A$ and all s . This proves (ii).

Theorem 25 of the source paper says that if a bounded set A countably supports the induced subspace $D(\mathfrak{s})$ of a projectional skeleton, then there is a SPRI $(Q_\alpha)_{\alpha \leq \kappa}$ with $Q_\alpha a \in \{0, a\}$ for every $a \in A$ and every $\alpha \leq \kappa$. Our set A countably supports all of X^* , hence certainly $D(\mathfrak{s})$, so (ii) implies (iii).

Finally assume either (ii) or (iii). The skeleton or SPRI invariance is not needed for this reverse implication; the ℓ_2 -generator alone reconstructs a Hilbert generator. Let $H = \ell_2(A)$ with canonical orthonormal basis $(e_a)_{a \in A}$. Define S on finitely supported vectors by

$$S \left(\sum_{a \in F} c_a e_a \right) = \sum_{a \in F} c_a a.$$

Using (1), for every finite scalar family $(c_a)_{a \in F}$,

$$\left\| \sum_{a \in F} c_a a \right\| = \sup_{\|x^*\| \leq 1} \left| \sum_{a \in F} c_a x^*(a) \right| \leq \left(\sum_{a \in F} |c_a|^2 \right)^{1/2} \sup_{\|x^*\| \leq 1} \left(\sum_{a \in F} |x^*(a)|^2 \right)^{1/2} \leq C \left(\sum_{a \in F} |c_a|^2 \right)^{1/2}.$$

Hence S extends to a bounded operator $\ell_2(A) \rightarrow X$. Since $\overline{\text{span}}(A) = X$, the range of S is dense. Therefore X is Hilbert generated. $\square$

Verification Notes

The argument uses only:

- the standard operator definition of Hilbert-generated spaces;
- the source paper’s quoted equivalence between Plichko spaces and commutative projectional skeletons;
- Corollary 21 of the source paper, which makes a skeleton invariant on a countably supporting set;
- Theorem 25 of the source paper, which constructs the invariant SPRI.

The separable case is not the intended target of Problem 34. In any event, every separable Banach space is Hilbert generated by a bounded operator from ℓ_2 with dense range, built from a norm-dense sequence.

Limitations and Novelty Check

This packet gives a full formal equivalence, but the Hilbert information is encoded in the explicit ℓ_2 -generator estimate. It should be read as a minimal skeleton/SPRI characterization, not as a claim that no more intrinsic “shrinkingness” formulation exists.

The run indexes were searched for arXiv:2110.02320 and for the phrases “Hilbert generated”, “projectional skeleton”, “SPRI”, and “Markushevich”. No existing packet for Problem 34 or this invariant ℓ_2 -generator reformulation was found. A bounded web search for exact Hilbert-generated skeleton/SPRI formulations did not identify an exact prior answer.

References

1. C. Correa, M. Cuth, and J. Somaglia, *Characterizations of weakly $\mathcal{K}$ -analytic and Vasak spaces using projectional skeletons and separable PRI*, arXiv:2110.02320.
2. M. Fabian, G. Godefroy, P. Hajek, and V. Zizler, *Hilbert-generated spaces*, J. Funct. Anal. 200 (2003), 301–323.